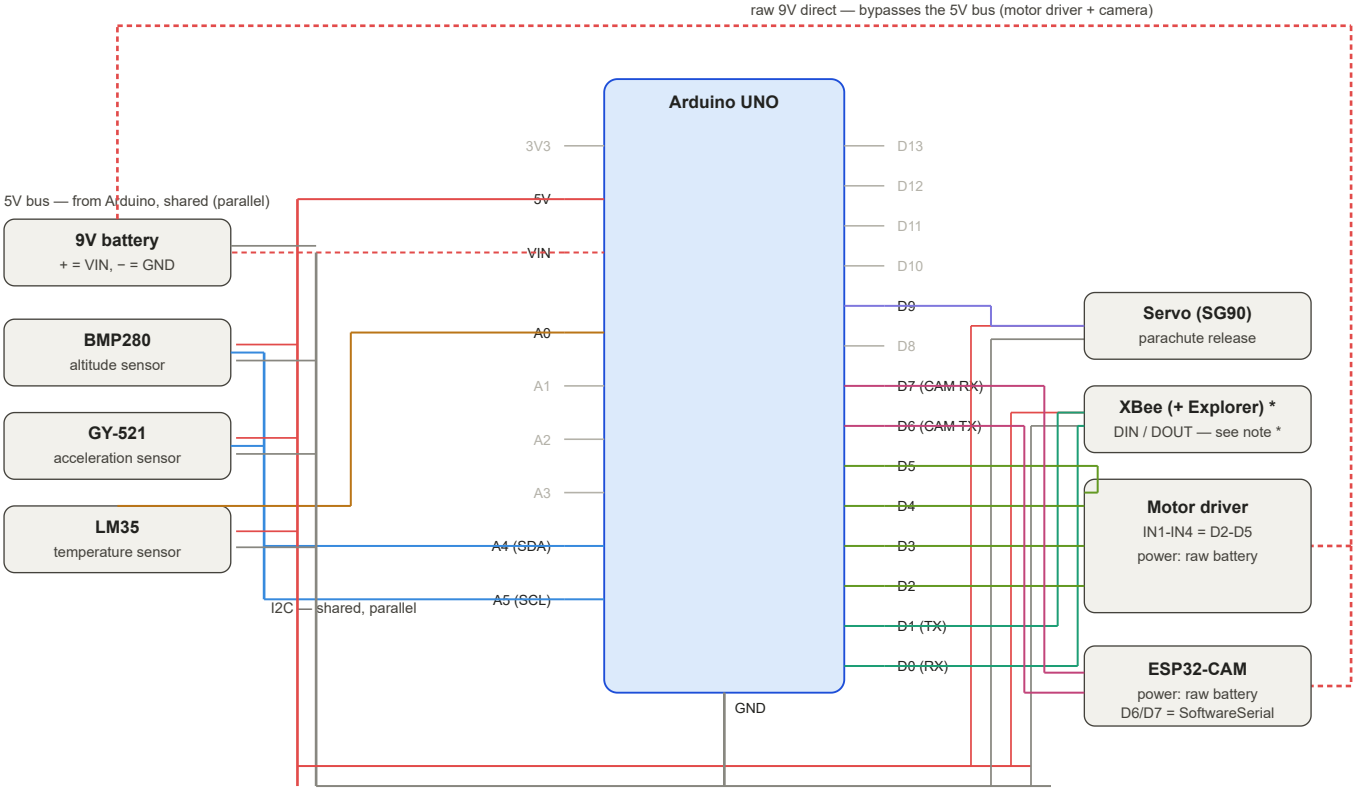




Legend

Power & Ground

- Solid red = 5V bus (shared, parallel) · Dashed red = raw 9V direct from battery
- Gray = GND — common return, shared by everything

Signal Wires

- Blue = I2C bus (BMP280 + GY-521 share A4/A5, in parallel)
 - Orange = LM35 analog signal (A0, dedicated wire)
 - Purple = servo signal (D9, dedicated wire)
 - Green = motor driver control (D2–D5, dedicated wires)
 - Teal = XBee TX/RX (D0/D1 — disconnect before uploading code)
 - Magenta = ESP32-CAM serial (D6/D7 via SoftwareSerial, avoids the XBee's D0/D1)
- Gray pin labels (3V3, A1–A3, D8, D10–D13) = unused pins, no wire connected

I2C Address Configuration

BMP280: CSB → Arduino 5V, SDO → Arduino GND (sets address 0x76)
GY-521: AD0 → Arduino GND (sets address 0x68)

Design Notes & Cautions

* Verify your XBee Explorer regulates 5V→3.3V before powering; use Arduino 3V3 if unsure
Motor driver ENA/ENB ship jumpered to 5V by default (full speed, no PWM needed)
Motor driver & ESP32-CAM use raw battery power to avoid current spikes disturbing sensors
Lines crossing without a junction dot are NOT connected
All grounds (GND) are common across the whole circuit